Smart inventory and warehouse management system with environmental monitoring

Software Engineering Principles Project

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Student information :

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Lab 1 – Problem Definition

Problem Description :

Warehouses play a crucial role in storing goods for hospitals, supermarkets, pharmaceutical companies, and retail businesses. Traditional inventory management focuses mainly on tracking stock levels, but it often neglects the environmental conditions required for specific products. For example, medicines and vaccines must be stored at a certain temperature, food products may require refrigeration, and electronics need protection from high humidity. The Smart Inventory & Warehouse Management System with Environmental Monitoring not only manages stock entry, exit, and availability but also monitors environmental conditions inside the warehouse using IoT sensors (temperature, humidity, etc.). The system ensures that each type of product is stored under its required conditions. In case of any abnormal changes, automatic alerts are generated for administrators to take corrective action. This reduces losses, ensures product safety, and increases efficiency.

Lab 2 – Information Gathering

Interviews :

1 –

Name of Interviewee : Mr. Anas Alshaikh,

Job title : Warehouse Manager - Medical Supplies at Fakeeh Hospital

Interview date: 7-9-2025

- Do you currently use a digital system to manage the warehouse?
- Is monitoring the temperature important in your warehouse?
- Do you need automatic alerts when stock levels are low ?
- What are the main challenges you face in managing the warehouse.
- How important is it to include environmental monitoring (temperature, humidity) in the system ?

Interview concusion : From the interview it is clear that the warehouse does not have a system for monitoring environmental conditions, which leads to risks such as damaged or expired medical supplies.

The warehouse needs a new automated management system that provides the following :

1. **Real time** tracking of inventory
2. **Enviromemtal monitoring** (temperature and humidity)
3. **Automatic alerts** for low stock and expiry date

2 –

Name of Interviewee: Mr. Fahad Alqahtani
Job Title : Warehouse Staff at Aramco Interview
Date: 10/09/2025

- Can you describe your role and daily responsibilities in the warehouse?
- Do you currently use any digital system for tracking stock levels?
- What features would you like a smart inventory system to provide in order to make your work easier ?
- Would you find it useful to receive SMS or email alerts when environmental conditions go out of range ?
- What difficulties do you face in monitoring temperature and humidity for sensitive products?

Interview conclusion : The current inventory management system still relies heavily on manual processes, which often cause errors and delays in updating data.
The warehouse needs a new automated management system that provides the following :

1. **Real-time inventory tracking** to reduce errors and delays caused by manual processes.
2. **Continuous environmental monitoring** of temperature and humidity to protect sensitive products
3. **Instant alerts (SMS/Email/Notification)** when conditions deviate from the required range
4. **Centralized dashboard** to provide administrators with a clear view of stock levels and warehouse conditions**.**

Lab 3 – planning phase 1

Purpose of the project

Business Value and Importance of the Project

1. Reducing Financial Losses

The system minimizes product damage caused by unsuitable storage conditions, especially for sensitive items such as medicines and food products

2. Increasing Operational Efficiency:

Automation of stock tracking saves time and reduces manual effort, leading to faster and more accurate warehouse operations.

3. Supporting Better Decision-Making:

Real-time data and reporting help managers make informed decisions and respond quickly to any issues

. 4. Improving Customer Satisfaction:

Accurate inventory tracking ensures product availability and quality, enhancing customer trust and satisfaction.

5. Encouraging Digital Transformation:

The project supports the organization's digital transformation by integrating modern IoT technologies for smarter warehouse management.

main motivation for this project is to address real-world challenges in warehouse management by providing a reliable and intelligent automated solution

The current system's limitations, such as manual processes and lack of monitoring, highlight the need for a smart, automated, and data-driven solution

Goals of the Project

The system aims to replace manual inventory management with an automated, real-time solution that enhances efficiency, reduces human errors, and prevents product loss or damage.

Preliminary Report

1. The Problem

The current warehouse system is mostly manual, which causes errors, delays, and inaccurate inventory data. There is no monitoring of temperature or humidity, which can lead to product damage, especially for sensitive items like medicines and food.

2. Findings

Interviews showed that warehouses lack digital systems and environmental monitoring. Staff still rely on manual work, and there are no alerts for low stock or unsafe conditions. These issues reduce efficiency and increase product loss.

3. Proposed Solution

The proposed solution is to develop a Smart Inventory and Warehouse Management System with Environmental Monitoring. The system will use IoT sensors to track temperature and humidity, update stock automatically, and send alerts when problems occur. This will improve accuracy, save time, and protect stored products.

Lab 4 – planning phase 2

Feasibility Study

Executive Summary:

The Smart Inventory & Warehouse Management System with Environmental Monitoring project is feasible from all angles technically, operationally, financially, and within the planned schedule. It effectively addresses the need for smarter, automated warehouse systems capable of monitoring temperature and humidity to ensure proper storage for sensitive goods like food, pharmaceuticals, and electronics. After evaluating costs, available technology, and market demand, the project is deemed fully feasible and recommended for development.

Project description:

Scope of the Project

Covers:

- Implementation of an IoT-based warehouse management system in selected warehouses within the company network.
- Real-time environmental data monitoring (temperature, humidity). • Automated alerts and reporting for abnormal conditions.

Does not cover:

- Warehouses without internet connectivity.
- Manual inventory systems or external supply chain integration.
- Global deployment (limited to local/regional warehouses).

Time scope:

- The system will be implemented and tested within one academic semester (approx. 4 months).
- Includes phases: planning, implementation, testing, and maintenance

Market Feasibility:

There is a strong market need for this project. Many businesses (e.g., hospitals, supermarkets, and pharmacies) struggle with inventory loss due to environmental mismanagement. This system provides an automated, efficient solution that improves safety, reduces losses, and aligns with modern smart warehouse trends, making it highly marketable and competitive.

Technical Feasibility:

The technology required: IoT sensors, cloud-based data storage, and management software is readily available and affordable. The team has the technical expertise in database design, web development, and IoT integration to build the system. Therefore, the system is technically feasible with minimal risk of technology failure.

Organizational Feasibility:

The system can be easily integrated into existing warehouse operations. It supports standard inventory workflows (entry, exit, tracking) while adding automated monitoring and alert functions. Users (admins and staff) can learn and adapt quickly through an intuitive dashboard and QR code scanning feature. Hence, it is operationally feasible within the current organizational setup.

Financial Feasibility:

The project requires minimal financial investment since most resources are digital (software tools, IoT kits).

Estimated costs:

Resource	Quantity	Cost
Iot Sensor Kit(temperature and humidity)	3	1500 SAR
Server / cloud hosting	1	450 SAR
Development tools (IDE,DB,etc)	-	0 (free / open source)
Developer	2	13,000 SAR
Tester	1	10,000 SAR
Manager	4	0 (student work)
Total		24,950 SAR

Schedule feasibility:

The project fits within the semester timeframe (4 months), including planning, development, testing, and deployment. Thus, both financially and time-wise, it is feasible.

Risk Analysis:

Risk	**Type**	**Impact**	**Action required**
Sensor failure	Technical	Medium	Use backup sensors and regular calibration
Network downtime	Technical	High	Local data catching until connection restored
Data loss	Operational	High	Cloud backup and daily logs
Budget overrun	Financial	Low	Use open source tools and student resources
Delays in implementation	Schedule	Medium	Weekly progress tracking and task division

All risks have preventive and corrective measures to ensure smooth progress.

Project plan:

Phases and activities

Phase	Activity	Duration	Resources	Cost
1) **Planning**	Requirement gathering feasibility study	1 week	Team members	-
2) **System design**	Database ,architecture ,UI design	1 week	Developers	-
3) **Implementation**	Bancked an IoT integration	3 weeks	Developers and IoT kits	15000 SAR
4) **Testing**	Functional performance ,sensor accuracy	1 week	Testers	-
5) **Deployment**	Install and run system in warehouse	1 Week	All members	450 SAR
6) **Maintenance**	Fix bugs ,optimize system	Ongoing	Developers	-

Total duration: approximatly 7 weeks

Total estimated cost: approximatly 1,950 SAR

Project Lifecycle:

Planning → Design → Implementation → Testing → Deployment → Maintenance.

Lab 5 – Analysis phase

- **Stakeholders List :**

Warehouse Manager : Main user of the system. Responsible for monitoring inventory levels, environmental conditions, and receiving alerts.

Warehouse Staff : Use the system to record stock entry and exit, update product information, and respond to alerts

System Administrator : Manages system settings, user accounts, and ensures system performance and data security.

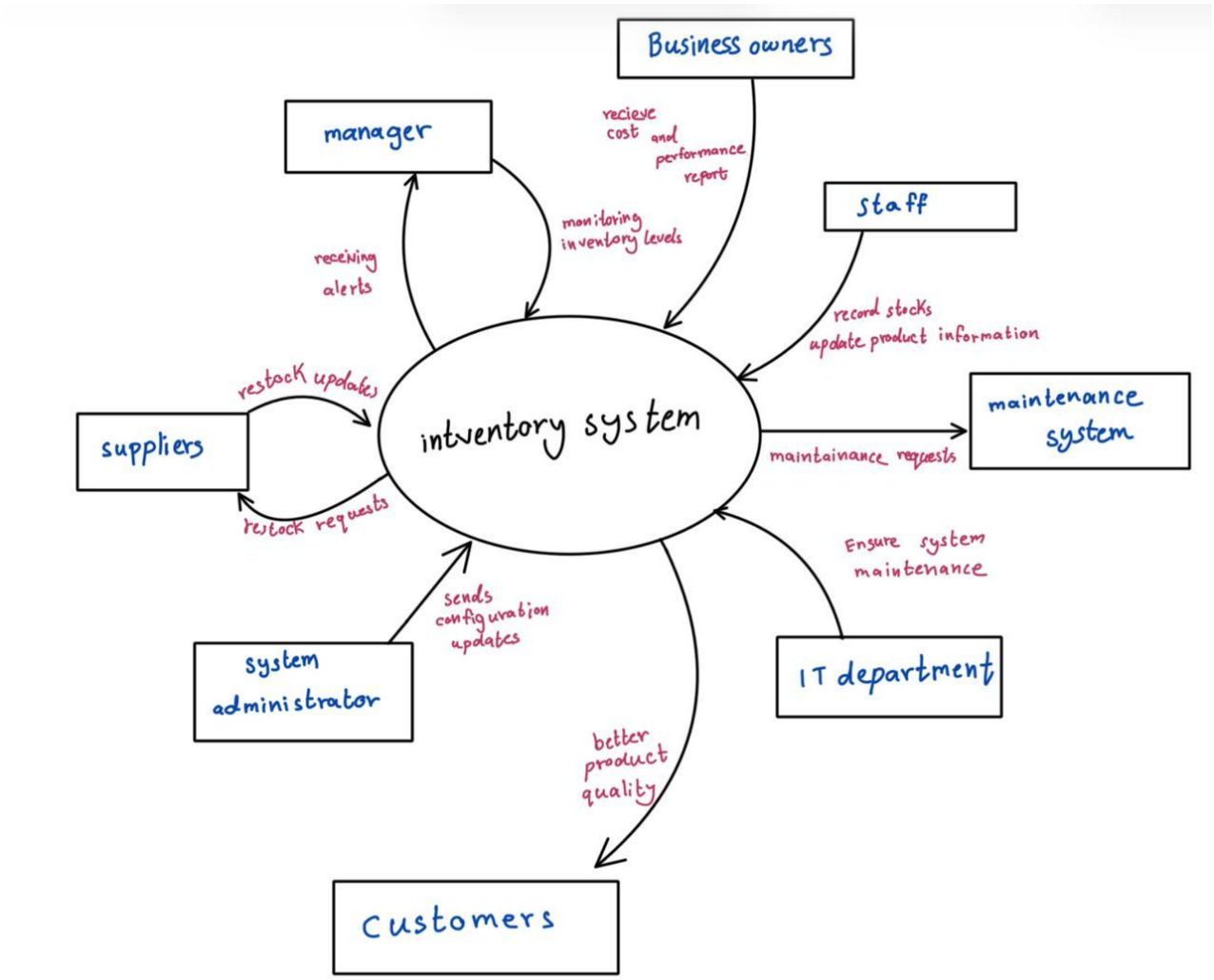
IT Department : Responsible for developing, testing, and maintaining the system. Suppliers : Benefit from accurate inventory updates to know when restocking is needed.

Customers: Indirect stakeholders; benefit from better product quality and availability.

Company Management / Business Owners : Interested in cost reduction, efficiency improvement, and decision-making based on real-time data.

Maintenance Team : Ensures that sensors and devices are functioning properly.

- **Context diagram :**



- **Inventory System Event Table**

Event(Trigger)	Source (external entity)	Response (process)	Destination (stakeholders)
New stock arrives stock is moved out	Warehouse staff	Record stock entry exit and update inventory levels	Warehouse manager ,company management ,suppliers
Product information needs updates	Warehouse staff	Update product information in the system	Warehouse Manager ,Company Management
Inventory level needs monitoring	Internal (system)	Monitor inventory levels continuously	warehouse managers
Inventory level drops below threshold	Internal (system)	Generate low stock alerts and send Restock requests	Warehouse manager ,suppliers
Environmental condition needs monitoring	Internal (system)	Monitor environmental data temperature humidity etc	Warehouse manager
Requests for cost /performance reports	Business owners /Company management	Generate cost and performance report	Business owners /management
Restock update is received	Suppliers	Update inventory levels and restock status	Warehouse manager ,company Management
Configuration settings need modification	System Administrator	Send configuration updates	System Administrator ,IT Department
System is due for maintenance	IT department	Ensure system maintenance and reliability	IT department ,System administrator
Sensor/ device malfunctioning	Maintenance team	Generate maintenance requests	Maintenance team
Receiving an alert	Warehouse manager	Receive alert system pushes alert data	Warehouse manager

Lab 6 – Requirements specification document

Functional Requirements
for Smart Inventory System

FR1 – Manage Inventory Records

- FR1.1: The system shall allow staff to add new products to the inventory.
- FR1.2: The system shall allow staff to update product quantity, location, and status.
- FR1.3: The system shall automatically reduce the stock when items are issued.

FR2 – Environmental Monitoring

- FR2.1: The system shall collect temperature and humidity data from IoT sensors.
- FR2.2: The system shall display real-time environmental readings on the dashboard.

FR3 – Alerts & Notifications

- FR3.1: The system shall send alerts when temperature or humidity goes out of the safe range.
- FR3.2: The system shall notify the manager when stock level is below the minimum threshold.

FR4 – User Authentication

- FR4.1: The system shall allow users to log in using a valid username and password.
- FR4.2: The system shall restrict access based on user roles (manager, staff, admin).
-

FR5 – Reports Generation

- FR5.1: The system shall generate inventory status reports.
- FR5.2: The system shall generate environmental condition reports.
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FR6 – Track Product Expiry

- FR6.1: The system shall alert users when a product is close to its expiry date.

Non-Functional Requirements

NFR1 – Performance

- NFR1.1: The system shall update sensor readings every 5 seconds.
- NFR1.2: The system shall load inventory data within 2 seconds.

NFR2 – Reliability

- NFR2.1: The system shall operate 24/7 with minimal downtime.
- NFR2.2: The system shall store sensor data even if the network disconnects temporarily.

NFR3 – Usability

- NFR3.1: The system shall provide a simple, user-friendly interface.
- NFR3.2: Users shall be able to learn the system within one training session.

NFR4 – Security

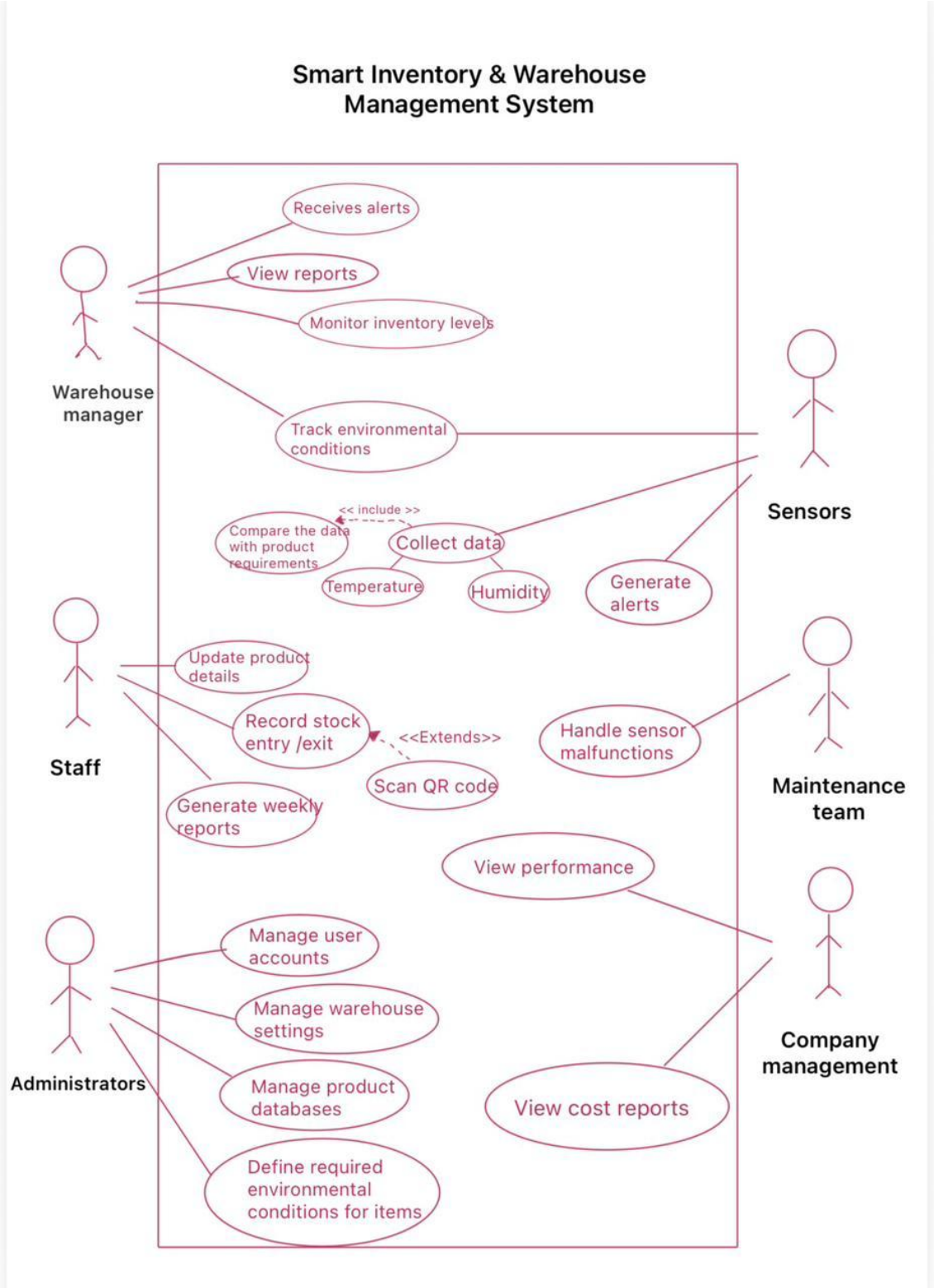
- NFR4.1: The system shall encrypt all user login information.
- NFR4.2: Only authorized users shall have access to inventory updates and sensor data.

NFR5 – Compatibility

- NFR5.1: The system shall be accessible on web browsers and mobile devices.

Lab 7 – Analysis phase 2

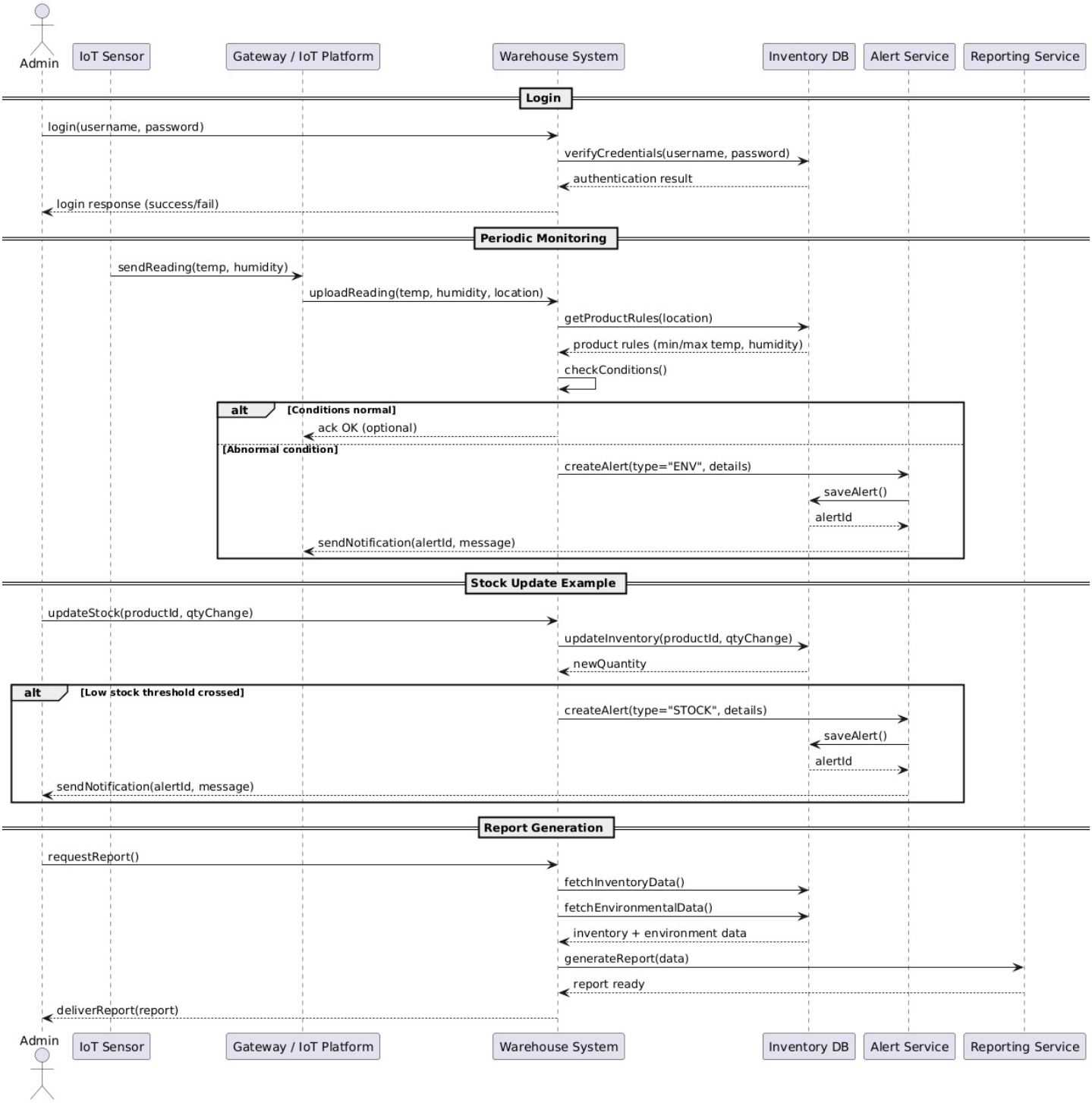
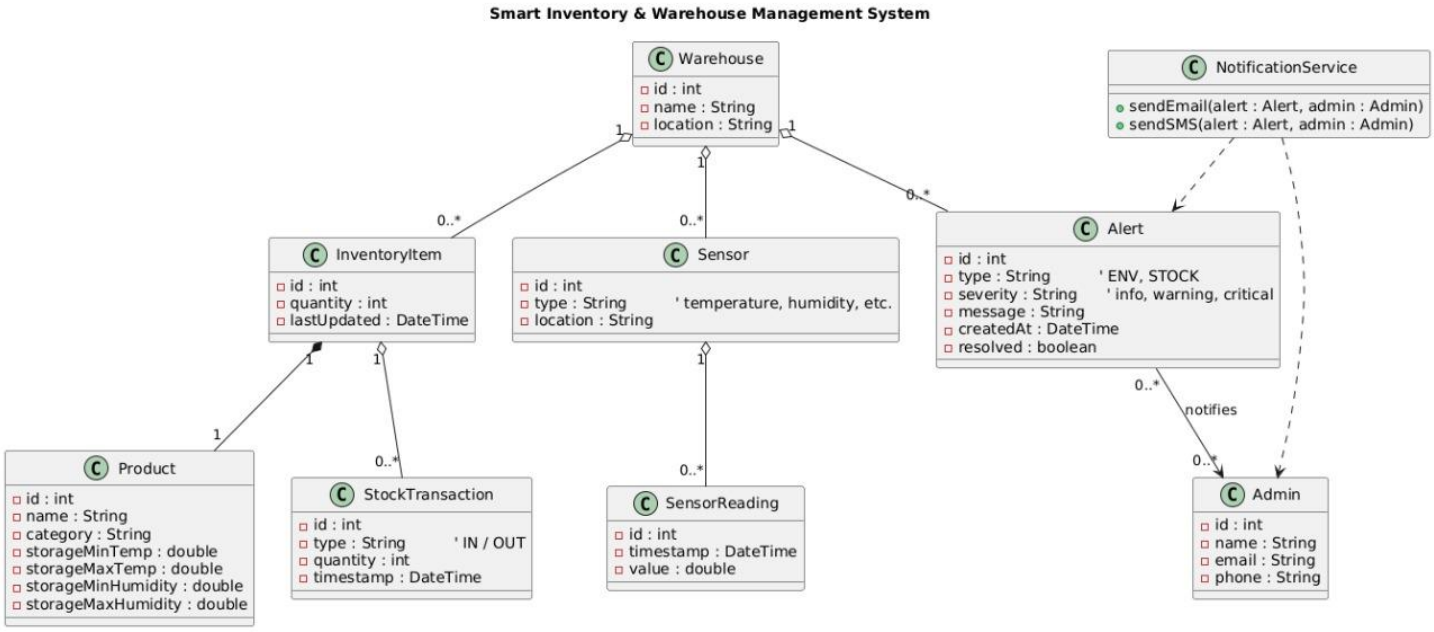
Use case diagram :



Use case description :

The Smart Inventory & Warehouse Management System is used by Warehouse Managers, Warehouse Staff, System Administrators, Maintenance Teams, and Company Management. The system manages product information, warehouse configurations, inventory transactions, and real time environmental data collected through IoT sensors, while also sending low-stock notifications to the appropriate users. Warehouse Staff record stock entry and exit, update product details, generate weekly reports, and scan QR codes for faster processing. Warehouse Managers monitor inventory levels, view system reports, track environmental conditions, and receive low-stock notifications whenever item quantities drop below the required threshold. System Administrators manage user accounts, warehouse settings, product databases, and define the required environmental conditions for stored items. Sensors continuously collect temperature and humidity data, which the system stores and displays for monitoring. The system also tracks inventory quantities and automatically generates low-stock notifications for the Warehouse Manager. The Maintenance Team handles sensor malfunctions through maintenance requests generated by the system, while Company Management uses the system to view cost and performance reports.

Lab 8 – Analysis phase 3



Lab 9 – Modeling phase